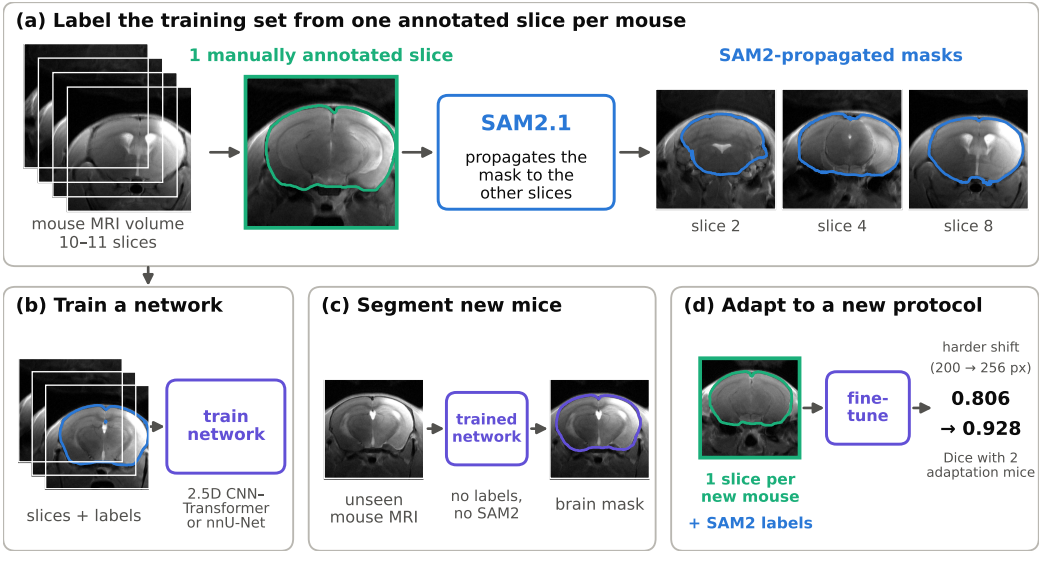

(a) Label the training set from one annotated slice per mouse
1 manually annotated slice
SAM2-propagated masks
SAM2.1
propagates the mask to the other slices
mouse MRI volume
10–11 slices
slice 2
slice 4
slice 8
(b) Train a network
train network
slices + labels
2.5D CNN–Transformer or nnU-Net
(c) Segment new mice
trained network
unseen mouse MRI
no labels, no SAM2
brain mask
(d) Adapt to a new protocol
fine-tune
harder shift
(200 → 256 px)
0.806
→ 0.928
Dice with 2 adaptation mice
1 slice per new mouse
+ SAM2 labels